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# SSC-I NATIONAL BOOK FOUNDATION FBISE

## MATHEMATICS TEST - CHAPTER 2: LOGARITHMS

| Name: | | Roll No. | | Section: |
|---|---|---|---|---|

**Total Time: 2:30 Hour** **Total Marks: 150**
**Time Allowed: 150 Minutes**

**PAPER OBJECTIVE**
**SECTION-A (MARKS 150)**

NOTE: Section-A is compulsory. All parts of this section are to be answered on these pages and handed over to the center superintendent. Deleting / overwriting is not allowed. Do not use lead pencil.

**Q.1 Fill the relevant bubble for each part. All parts carry one mark.**

| Questions | A | B | C | D | A B C D |
|---|---|---|---|---|---|
| 1. The mass of Uranus (87 trillion trillion kg) in scientific notation is | $8.7 \times 10^{24}$ kg | $8.7 \times 10^{25}$ kg | $87 \times 10^{25}$ kg | $8.7 \times 10^{26}$ kg | ○●○○ |
| 2. The position after the first left-hand non-zero digit is called the | Mantissa | Reference position | Characteristic | Base | ○●○○ |
| 3. When the decimal point is moved to the right, the value of n is | Positive | Zero | One | Negative | ○○○● |
| 4. One light year in scientific notation is about | $5.88 \times 10^{9}$ miles | $5.88 \times 10^{-12}$ miles | $5.88 \times 10^{12}$ miles | $58.8 \times 10^{12}$ miles | ○○●○ |
| 5. In scientific notation, the decimal point is placed after the | Last digit | First non-zero digit on the left | Second digit | First zero | ○●○○ |
| 6. If a number is smaller than 1, the exponent in scientific notation is | Positive | Zero | Negative | One | ○○●○ |
| 7. Sunlight takes about how long to reach the Earth? | 5 min | 1 hour | 2 min 30 s | 8 min 20 s | ○○○● |
| 8. When the decimal point is moved to the left, the value of n is | Positive | Negative | Zero | One | ●○○○ |
| 9. On most calculators, a number in scientific notation is entered using the key | LOG | E or EXP | ANS | MODE | ○●○○ |
| 10. A number c is in scientific notation if $c = d \times 10^{n}$ where | $0 \le d < 10$ and $n \in N$ | $1 \le d \le 10$ and $n \in N$ | $0 < d < 1$ and $n \in Z$ | $1 \le d < 10$ and $n \in Z$ | ○○○● |
| 11. If a number is greater than 10, the exponent in scientific notation is | Positive | Negative | Zero | Undefined | ●○○○ |
| 12. Write $6.1 \times 10^{4}$ in standard notation | 6100 | 61000 | 610 | 610000 | ○●○○ |
| 13. Write $6.53 \times 10^{-6}$ in standard notation | 0.000653 | 0.0000000653 | 0.0000653 | 0.00000653 | ○○○● |
| 14. Write 53400000 in scientific notation | $5.34 \times 10^{6}$ | $5.34 \times 10^{7}$ | $5.34 \times 10^{8}$ | $53.4 \times 10^{6}$ | ○●○○ |



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| Questions | A | B | C | D | A B C D |
|---|---|---|---|---|---|
| **15. Simplify $(9 \times 10^8) \div (3 \times 10^{-2})$ in scientific notation** | $6 \times 10^{10}$ | $3 \times 10^{10}$ | $3 \times 10^9$ | $3 \times 10^6$ | ○●○○ |
| **16. Write $1.2 \times 10^5$ in standard notation** | 12000 | 1200 | 1200000 | 120000 | ○○○● |
| **17. Write 8300000 in scientific notation** | $83 \times 10^5$ | $8.3 \times 10^4$ | $8.3 \times 10^6$ | $8.3 \times 10^5$ | ○○●○ |
| **18. Write 310000000 in scientific notation** | $3.1 \times 10^{-8}$ | $3.1 \times 10^8$ | $3.1 \times 10^6$ | $3.1 \times 10^9$ | ○●○○ |
| **19. Write 0.0032 in scientific notation** | $3.2 \times 10^{-2}$ | $3.2 \times 10^{-3}$ | $3.2 \times 10^{-5}$ | $3.2 \times 10^{-1}$ | ○●○○ |
| **20. Write $2.4 \times 10^{-3}$ in standard notation** | 0.0024 | 0.24 | 0.000024 | 0.024 | ●○○○ |
| **21. Write 0.00053407 in scientific notation** | $5.3407 \times 10^{-4}$ | $5.3407 \times 10^{-2}$ | $5.3407 \times 10^{-6}$ | $5.3407 \times 10^{-5}$ | ●○○○ |
| **22. The logarithm of zero is** | Zero | One | Not defined | Negative | ○○●○ |
| **23. The inverse of an exponential relation is a** | Linear relation | Quadratic relation | Radical relation | Logarithmic relation | ○○○● |
| **24. Log tables were first developed in the late 1500s by** | Henry Briggs | John Napier | Al-Khwarizmi | Pythagoras | ○●○○ |
| **25. The word "logarithm" is derived from the name of** | Napier | Briggs | Al-Khwarizmi | Euclid | ○○●○ |
| **26. The logarithmic form of $b^y = x$ is** | $y = \log_b x$ | $x = \log_b y$ | $b = \log_y x$ | $y = \log_x b$ | ●○○○ |
| **27. A relation of the form $x = b^y$ $(b > 0, b \neq 1, x > 0)$ is called an** | Exponential relation | Logarithmic relation | Linear relation | Quadratic relation | ●○○○ |
| **28. $\log_b (b^x)$ equals** | b | x | $b^x$ | 1 | ○●○○ |
| **29. If $b^y = x$ $(b > 0, b \neq 1)$, then y is the** | Antilog of x | Characteristic of x | Logarithm of x to base b | Mantissa of x | ○○●○ |
| **30. The logarithm of a negative number is** | Negative | Zero | Not defined | One | ○○●○ |
| **31. The base of a logarithm cannot be** | 1 | 2 | 10 | e | ●○○○ |



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| Questions | A | B | C | D | A B C D |
|---|---|---|---|---|---|
| **32. log_b b equals** | 0 | b | 1 | −1 | ○○●○ |
| **33. log_b x is defined only for** | Negative x | All real x | Zero | Positive x | ○○○● |
| **34. If $\log_2 x = 4$, then x =** | 16 | 8 | 18 | 15 | ●○○○ |
| **35. Find the value of $\log_2 1$** | 2 | −3 | 1 | 0 | ○○○● |
| **36. Find the value of $\log_{49} 7$** | 1/2 | −1 | 4 | 3/2 | ●○○○ |
| **37. If log_x 1000 = 3, then x =** | 10 | 333 | 11 | 20 | ●○○○ |
| **38. Find the value of $\log_8 2$** | 2/3 | 1/3 | −1/3 | 4/3 | ○●○○ |
| **39. If $\log_2 (x - 1) = 3$, then x =** | 10 | 8 | 7 | 9 | ○○○● |
| **40. Find the value of $\log_2 1/8$** | 2 | 5 | −4 | −3 | ○○○● |
| **41. Find the value of ln $e^5$** | 5 | 5e | e | 0 | ●○○○ |
| **42. If log_x 32 = 5, then x =** | 5 | 2 | 4 | 3 | ○●○○ |
| **43. The exponential form of $\log_5 125 = 3$ is** | $3^5 = 125$ | $125^5 = 3$ | $5^{125} = 3$ | $5^3 = 125$ | ○○○● |
| **44. $\log_2 (4 - 2x) = y$ is defined for x = 2?** | Yes, y = 1 | Yes, y = 0 | No, $2^y = 0$ is impossible | Yes, y = 2 | ○○●○ |
| **45. The logarithmic form of $7^4 = 2401$ is** | $\log_{2401} 7 = 4$ | $\log_4 2401 = 7$ | $\log_7 2401 = 4$ | $\log_7 4 = 2401$ | ○○●○ |
| **46. The exponential form of $\log_{10} 100 = 2$ is** | $100^{10} = 2$ | $10^2 = 100$ | $2^{10} = 100$ | $100^2 = 10$ | ○●○○ |
| **47. Find the value of $\log_{100} 100$** | 1/2 | 5 | 2 | 1 | ○○○● |
| **48. If $\log_3 x = 4$, then x =** | 64 | 84 | 81 | 80 | ○○●○ |



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| Questions | A | B | C | D | A B C D |
|---|---|---|---|---|---|
| **49. Find the value of $\log_{11} 121$** | 3 | −2 | 2 | 6 | ○○●○ |
| **50. Find the value of $\log_{10} 10^{-3} + \log_{10} 10^{5}$** | 1 | 2 | 8 | −2 | ○●○○ |
| **51. If $\log_{64} x = -4/3$, then x =** | 1/16 | 1/256 | −256 | 1/64 | ○●○○ |
| **52. If $\log_{2} x = -3$, then x =** | 1/8 | 1/6 | −6 | 8 | ●○○○ |
| **53. Find the value of $\log_{10} 1000$** | 1 | 3 | −1/2 | 4 | ○●○○ |
| **54. The exponential form of $\log_{3} 81 = 4$ is** | $81^{3} = 4$ | $4^{3} = 81$ | $3^{4} = 81$ | $3^{81} = 4$ | ○○●○ |
| **55. Find the value of $\log_{3} 243$** | 5 | 4 | 2 | −1/3 | ●○○○ |
| **56. If $\log_{2} 64 = x + 1$, then x =** | 5 | 7 | 6 | 3 | ●○○○ |
| **57. Find the value of $\log_{5} 625$** | 8 | −3 | 1 | 4 | ○○○● |
| **58. If $\log_{3} (x + 1) = 2$, then x =** | 8 | 6 | 7 | 5 | ●○○○ |
| **59. If $\log_{5} x = 3$, then x =** | 130 | 124 | 243 | 125 | ○○○● |
| **60. Find the value of $\log_{5} 125$** | 3 | −1/3 | −1/2 | 1/4 | ●○○○ |
| **61. If $\log_x 256 = 4$, then x =** | 5 | 3 | 4 | 8 | ○○●○ |
| **62. Find the value of $\log_{4} 64$** | 3 | −3 | −1/3 | 6 | ●○○○ |
| **63. The logarithmic form of $2^{10} = 1024$ is** | $\log_{1024} 10 = 2$ | $\log_{2} 10 = 1024$ | $\log_{2} 1024 = 10$ | $\log_{1024} 2 = 10$ | ○○●○ |
| **64. If $\log (x + 3) = \log (15x - 4)$, then x =** | 1 | 2 | 7 | 0.5 | ○○○● |
| **65. Logarithms to base e are called** | Natural logarithms | Common logarithms | Decimal logarithms | Binary logarithms | ●○○○ |



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|---|---|---|---|---|---|
| **66. The logarithm of a number equals** | Characteristic − Mantissa | Characteristic + Mantissa | Characteristic × Mantissa | Characteristic ÷ Mantissa | ○●○○ |
| **67. ln 1 equals** | 1 | e | 0 | 10 | ○○●○ |
| **68. Logarithms to base 10 are called** | Natural logarithms | Binary logarithms | Common logarithms | Exponential logarithms | ○○●○ |
| **69. log 1 equals** | 1 | 0 | 10 | −1 | ○●○○ |
| **70. If 1 < x < 10, then log x lies between** | 1 and 2 | 0 and 1 | −1 and 0 | 10 and 100 | ○●○○ |
| **71. The mantissa is always** | Negative | An integer | Non-negative and less than 1 | Greater than 1 | ○○●○ |
| **72. The decimal (fractional) part of a logarithm is called the** | Mantissa | Characteristic | Antilog | Index | ●○○○ |
| **73. For a whole number, the number of digits equals** | Characteristic + 1 | Characteristic − 1 | Mantissa + 1 | Characteristic | ●○○○ |
| **74. The antilog of y is the number x such that** | x = log y | log x = y | x + y = 0 | x·y = 1 | ○●○○ |
| **75. The characteristic of the log of $3.3 \times 10^2$ is** | 3 | 2 | −2 | 1 | ○●○○ |
| **76. log $10^x$ equals** | 10 | 1 | 10x | x | ○○○● |
| **77. The characteristic of the log of a number between 1 and 10 is** | 1 | −1 | 0 | 10 | ○○●○ |
| **78. Common logarithms are also called** | Napier's logarithms | Briggs' logarithms | Euler logarithms | Binary logarithms | ○●○○ |
| **79. Natural logarithms are also called** | Briggs' logarithms | Common logarithms | Naperian logarithms | Decimal logarithms | ○○●○ |
| **80. If no base is written in log x, the base is understood to be** | 10 | e | 2 | 1 | ●○○○ |
| **81. The mantissa of a logarithm is found from the** | Antilog table | Log table | Sine table | Number line | ○●○○ |
| **82. The integral part of a logarithm is called the** | Mantissa | Antilog | Characteristic | Base | ○○●○ |



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|---|---|---|---|---|---|
| **83. ln ($e^x$) equals** | x | e | 1 | $e^x$ | ●○○○ |
| **84. log 10 equals** | 0 | 10 | e | 1 | ○○○● |
| **85. In an antilog table, the characteristic is used for** | Locating the decimal point | Finding the mantissa | Finding the base | Finding the index | ●○○○ |
| **86. The two most commonly used bases of logarithms are** | 2 and 3 | 1 and 10 | e and π | 10 and e | ○○○● |
| **87. The number e is a(n)** | Irrational number | Rational number | Integer | Whole number | ●○○○ |
| **88. ln e equals** | 0 | e | 10 | 1 | ○○○● |
| **89. The characteristic of the log of a number is the integral power of** | 2 | 10 | e | 1 | ○●○○ |
| **90. The inverse operation of taking a logarithm is** | Taking the mantissa | Taking the square root | Taking the antilog | Taking the base | ○○●○ |
| **91. The natural logarithm of x is denoted by** | ln x | log x | lg x | $e^x$ | ●○○○ |
| **92. The value of e is approximately** | 3.14159 | 1.41421 | 1.73205 | 2.71828 | ○○○● |
| **93. For finding an antilog, which part of the log is used to look in the table?** | Characteristic | Both parts | Mantissa | Neither | ○○●○ |
| **94. Common logarithms were introduced using base 10 by** | John Napier | Henry Briggs | Al-Khwarizmi | Descartes | ○●○○ |
| **95. The characteristic of log 0.537 is** | 0 | −3 | −2 | −1 | ○○○● |
| **96. Antilog of 0 equals** | 10 | 0 | Undefined | 1 | ○○○● |
| **97. The number of digits in $23^{15}$ is** | 21 | 23 | 22 | 19 | ●○○○ |
| **98. The characteristic of log 54 is** | −1 | 0 | 2 | 1 | ○○○● |
| **99. If log x = −2 + 0.0000, then x =** | 0.01 | 0.1 | 100 | 0.001 | ●○○○ |


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| Questions | A | B | C | D | A B C D |
|---|---|---|---|---|---|
| **100. The number of digits in the antilog of 5.3010 (whole number) is** | 4 | 3 | 5 | 6 | ○○○● |
| **101. The characteristic of log 4580 is** | 1 | 4 | 2 | 3 | ○○○● |
| **102. The number of digits in $5^{50}$ is** | 35 | 33 | 36 | 37 | ●○○○ |
| **103. The characteristic of log 12345 is** | −4 | 4 | 5 | 3 | ○●○○ |
| **104. The characteristic of log 5313 is** | 2 | 4 | −3 | 3 | ○○○● |
| **105. Product law of logarithms: log_b (mn) equals** | log_b m − log_b n | log_b m · log_b n | log_b m + log_b n | m log_b n | ○○●○ |
| **106. Before the invention of calculators, laws of logarithms were used on the** | Abacus only | Compass | Protractor | Slide rule | ○○○● |
| **107. Power law of logarithms: log_b ($m^n$) equals** | m log_b n | log_b m + n | (log_b m)$^n$ | n log_b m | ○○○● |
| **108. log_b a × log_a b equals** | 1 | 0 | ab | a/b | ●○○○ |
| **109. log_b a · log_c b · log_a c equals** | 0 | abc | a + b + c | 1 | ○○○● |
| **110. Logarithm laws convert multiplication and division into** | Squares and roots | Addition and subtraction | Powers and indices | Fractions and decimals | ○●○○ |
| **111. Which of the following is a true statement?** | log (a + b) = log a + log b | log (a − b) = log a − log b | log (ab) = log a + log b | log (a/b) = log a / log b | ○○●○ |
| **112. Change of base law: log_b m equals** | log_a b / log_a m | log_a m · log_a b | log_a m / log_a b | log_a m − log_a b | ○○●○ |
| **113. Quotient law of logarithms: log_b (m/n) equals** | log_b m + log_b n | log_b m / log_b n | n log_b m | log_b m − log_b n | ○○○● |
| **114. The product and quotient laws of logarithms require that the logs are** | Negative | Defined | Equal | Integers | ○●○○ |
| **115. A logarithm is basically a(n)** | Base | Exponent | Radical | Coefficient | ○●○○ |
| **116. log_a m is also equal to** | log_b m · log_a b | log_b m / log_a b | log_b m + log_a b | log_b m − log_a b | ●○○○ |


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|---|---|---|---|---|---|
| **117. The change of base law and the quotient law are** | The same law | Both false | Applicable only to base e | Often confused but different | ○○○● |
| **118. log (a − b) is equal to** | log a − log b | None of these | log a / log b | log a + log b | ○●○○ |
| **119. Find $\log_{10} 100 + \log_{10} 1000$** | 4 | 1100 | 5 | 6 | ○○●○ |
| **120. If log 2 = 0.3010 and log 5 = 0.6990, then log 10 =** | 1 | 0.2104 | 0.9 | 0.3980 | ●○○○ |
| **121. Expand: log ($5p^2$)** | log 5 + 2 log p | 2 log 5p | 5 log $p^2$ | log 5 · 2 log p | ●○○○ |
| **122. Find $\log_2 4 + \log_2 8$** | 32 | 5 | 12 | 4 | ○●○○ |
| **123. If log 2 = 0.3010, log 3 = 0.4771, then log 12 =** | 1.0791 | 1.2042 | 0.7781 | 1.1791 | ●○○○ |
| **124. Find $\log_2 8 + \log_2 8$** | 5 | 64 | 6 | 16 | ○○●○ |
| **125. If log 3 = 0.4771, then log 9 =** | 0.4771 | 0.2386 | 0.9542 | 1.9542 | ○○●○ |
| **126. If log 2 = 0.3010, log 3 = 0.4771, log 5 = 0.6990, then log 360 =** | 2.5562 | 2.5552 | 2.4771 | 1.5562 | ●○○○ |
| **127. $\log_2 8$ in base 10 equals** | log 8 / log 2 | log (8/2) | log 8 × log 2 | log 8 − log 2 | ●○○○ |
| **128. Find $\log_2 4 + \log_2 4$** | 6 | 5 | 4 | 8 | ○○●○ |
| **129. Find $\log_{10} 10 + \log_{10} 10000$** | 10010 | 6 | 100000 | 5 | ○○○● |
| **130. Find $\log_2 128 - \log_2 4$** | 4 | 32 | 124 | 5 | ○○○● |
| **131. Find 5 $\log_2 2$** | 5 | 10 | 7 | 6 | ●○○○ |
| **132. Find $\log_2 16 + \log_2 16$** | 32 | 256 | 8 | 9 | ○○●○ |
| **133. Find 4 $\log_3 3$** | 3 | 7 | 4 | 12 | ○○●○ |



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| Questions | A | B | C | D | A B C D |
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| **134. The 2005 earthquake in Pakistan measured on the Richter scale** | 8.2 | 6.5 | 8.5 | 7.6 | ○○○● |
| **135. For a ≠ 1, log_a x is not defined when x is** | Positive | Negative | 2 | 10 | ○●○○ |
| **136. The China earthquake (8.2) compared with the Pakistan earthquake (7.6) was nearly** | 2 times stronger | 4 times stronger | 10 times stronger | 0.6 times stronger | ○●○○ |
| **137. The Richter scale for earthquakes is a** | Logarithmic scale | Linear scale | Quadratic scale | Cubic scale | ●○○○ |
| **138. $\log_1 2$ is** | 0 | 1 | Undefined | 2 | ○○●○ |
| **139. On the Richter scale, the magnitude M is given by** | $M = I / I_0$ | $M = I_0 \log I$ | $M = (I_0 / I)^2$ | $M = \log (I / I_0)$ | ○○○● |
| **140. The 1978 earthquake in China measured on the Richter scale** | 7.6 | 8.2 | 6.5 | 9.1 | ○●○○ |
| **141. log_x (7 − x) is defined for x = 6 because** | Base is 1 | $7 - 6 < 0$ | Base $6 > 0$ and $7 - 6 > 0$ | It is a negative base | ○○●○ |
| **142. If $\log_2 (4 - 2x) = y$ and x = 1, then y equals** | 2 | 0 | 1 | 4 | ○○●○ |
| **143. The base 1 cannot be used since 1 raised to any power** | Is 0 | Is negative | Is infinite | Is always 1 | ○○○● |
| **144. log_x (7 − x) is not defined for x ≥ 7 because** | The base is 1 | $7 - x \leq 0$ | The base is negative | Both are fine | ○●○○ |
| **145. Kansu 1920 earthquake was 8.5 and Tokyo 1923 was 7.8. The stronger was stronger by about** | 1.1 times | 10 times | 5 times | 2 times | ○○●○ |
| **146. Two earthquakes differ by 1 unit on the Richter scale. The ratio of their intensities is** | ≈ 2 times | ≈ 5 times | 10 times | 100 times | ○○●○ |
| **147. An earthquake is 1 unit stronger on the Richter scale. Its intensity is** | 10 times greater | 1 time greater | Same | 2 times greater | ●○○○ |
| **148. Two earthquakes differ by 0.3 units on the Richter scale. The ratio of their intensities is** | 10 times | ≈ 2 times | ≈ 4 times | 0.6 times | ○●○○ |
| **149. If $M = \log (I/I_0) = 3$, then I =** | $100\ I_0$ | $1000\ I_0$ | $30\ I_0$ | $10\ I_0$ | ○●○○ |
| **150. Two earthquakes differ by 0.48 units on the Richter scale. The ratio of their intensities is** | 10 times | ≈ 5 times | 100 times | ≈ 3 times | ○○○● |


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